

NCOS Mesa Vegetation: Wrangling Brainstorm

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1. Set up

Packages

```
# general use
library(tidyverse)
library(janitor)

# file organization
library(here)

# data visualization
library(naniar)
library(patchwork)
library(scales)
```

Data

```
veg <- read_csv(here("data", "raw", "veg.csv"))

metadata <- read_csv(here("data", "raw", "vp_veg_metadata.csv"))
```

2. Exploring data structure

```
glimpse(veg)
```

```
Rows: 27,036
Columns: 17
$ x1          <dbl> 58888, 1580, 58889, 25359, 25360, 58887, 79653, 5888~
$ date        <date> 2021-08-04, 2021-08-04, 2021-08-04, 2021-08-
04, 202~
$ monitors    <chr> "Oliver, Hayden", "Oliver, Hayden", "Oliver, Hayden"~
$ transect_name <chr> "BM-01", "BM-01", "BM-01", "BM-01", "BM-01", "BM-
01"~
$ vp_axis     <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ site        <chr> "ncos", "ncos", "ncos", "ncos", "ncos", "ncos", "nco~
$ transect_length <dbl> 30, 30, 30, 30, 30, 30, 30, 30, 30, 30, 30, 30, 30, ~
$ num_quad    <dbl> 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, ~
$ transect_distance <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 3, 3, 3, 3, 3, 3, 3, 3~
$ transect_side <chr> "Left", "Left", "Left", "Left", "Left", "Left", "Left", "Lef~
$ vp_zone     <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ quadrat_notes <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ percent_cover <dbl> 1, 10, 2, 1, 1, 13, 25, 18, 16, 22, 4, 33, 1, 1, 10,~
$ psoc        <chr> "Avena_fatua", "Bare Ground", "Bromus_diandrus", "Di~
$ cover_category <chr> "NON-NATIVE COVER", "BARE GROUND", "NON-NATIVE COVER~
$ habitat     <chr> "Seasonal Brackish Marsh", "Seasonal Brackish Marsh"~
$ year        <dbl> 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021~
```

```
glimpse(metadata)
```

```
Rows: 257
Columns: 4
$ year_pool   <chr> "2021_BM-01", "2021_FSP-01", "2021_GL-02", "2021_GL-
07",~
$ year        <dbl> 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 20~
$ transect_name <chr> "BM-01", "FSP-01", "GL-02", "GL-07", "GL-09", "GL-
10", "~
$ num_quad    <dbl> 11, 15, 11, 10, 11, 11, 11, 11, 11, 10, 9, 11, 11, 11, 1~
```

```
unique(veg)
```

```
# A tibble: 27,036 x 17
  x1 date      monitors      transect_name vp_axis site transect_length
```

```

      <dbl> <date>      <chr>          <chr>          <chr>    <chr>          <dbl>
1 58888 2021-08-04 Oliver, Hayden BM-01      <NA>    ncos            30
2  1580 2021-08-04 Oliver, Hayden BM-01      <NA>    ncos            30
3 58889 2021-08-04 Oliver, Hayden BM-01      <NA>    ncos            30
4 25359 2021-08-04 Oliver, Hayden BM-01      <NA>    ncos            30
5 25360 2021-08-04 Oliver, Hayden BM-01      <NA>    ncos            30
6 58887 2021-08-04 Oliver, Hayden BM-01      <NA>    ncos            30
7 79653 2021-08-04 Oliver, Hayden BM-01      <NA>    ncos            30
8 58886 2021-08-04 Oliver, Hayden BM-01      <NA>    ncos            30
9 93848 2021-08-04 Oliver, Hayden BM-01      <NA>    ncos            30
10 1581 2021-08-04 Oliver, Hayden BM-01      <NA>    ncos            30
# i 27,026 more rows
# i 10 more variables: num_quad <dbl>, transect_distance <dbl>,
#   transect_side <chr>, vp_zone <chr>, quadrat_notes <chr>,
#   percent_cover <dbl>, psoc <chr>, cover_category <chr>, habitat <chr>,
#   year <dbl>

```

```
unique(veg$transect_name)
```

```

[1] "BM-01"  "FSP-01" "GL-02"  "GL-07"  "GL-09"  "GL-10"  "GL-11"
[8] "GL-12"  "GL-14"  "GL-15"  "PU-01"  "PU-02"  "PU-03"  "PU-04"
[15] "PU-05"  "PU-06"  "PU-07"  "SA-01"  "SF-01"  "SF-02"  "SML-01"
[22] "SML-02" "SML-03" "SML-04" "SML-05" "SML-06" "SML-07" "SMM-01"
[29] "SMM-02" "SMM-03" "SMM-04" "SMM-05" "SMM-06" "SMM-07" "SMR-01"
[36] "SMR-02" "SMT-01" "SMT-02" "SMT-05" "SMT-06" "SMT-07" "SMTH-18"
[43] "SMTH-21" "SMTH-22" "VP-01"  "VP-02"  "VP-03"  "VP-04"  "VP-05"
[50] "VP-06"  "VP-07"  "VP-08"

```

```
unique(veg$year)
```

```
[1] 2021 2022 2023 2024 2025
```

```
unique(veg$cover_category)
```

```

[1] "NON-NATIVE COVER" "BARE GROUND"      "NATIVE COVER"     "THATCH"
[5] "OTHER COVER"

```

```

veg |>
  distinct(transect_name, habitat) |>
  arrange(transect_name) |>
  print(n = Inf)

```

A tibble: 52 x 2

	transect_name	habitat
	<chr>	<chr>
1	BM-01	Seasonal Brackish Marsh
2	FSP-01	Seasonal Freshwater Pond
3	GL-02	Perennial Grassland
4	GL-07	Perennial Grassland
5	GL-09	Perennial Grassland
6	GL-10	Perennial Grassland
7	GL-11	Perennial Grassland
8	GL-12	Perennial Grassland
9	GL-14	Perennial Grassland
10	GL-15	Perennial Grassland
11	PU-01	Peripheral Uplands
12	PU-02	Peripheral Uplands
13	PU-03	Peripheral Uplands
14	PU-04	Peripheral Uplands
15	PU-05	Peripheral Uplands
16	PU-06	Peripheral Uplands
17	PU-07	Peripheral Uplands
18	SA-01	Sandy Annuals
19	SF-01	Sand Flat
20	SF-02	Sand Flat
21	SML-01	Salt Marsh
22	SML-02	Salt Marsh
23	SML-03	Salt Marsh
24	SML-04	Salt Marsh
25	SML-05	Salt Marsh
26	SML-06	Salt Marsh
27	SML-07	Salt Marsh
28	SMM-01	Salt Marsh
29	SMM-02	Salt Marsh
30	SMM-03	Salt Marsh
31	SMM-04	Salt Marsh
32	SMM-05	Salt Marsh
33	SMM-06	Salt Marsh
34	SMM-07	Salt Marsh
35	SMR-01	Remnant Salt Marsh
36	SMR-02	Remnant Salt Marsh
37	SMT-01	Transition - High Salt Marsh
38	SMT-02	Transition - High Salt Marsh
39	SMT-05	Transition - High Salt Marsh
40	SMT-06	Transition - High Salt Marsh

41	SMT-07	Transition - High Salt Marsh
42	SMTH-18	Transition - High Salt Marsh
43	SMTH-21	Transition - High Salt Marsh
44	SMTH-22	Transition - High Salt Marsh
45	VP-01	Vernal Pool
46	VP-02	Vernal Pool
47	VP-03	Vernal Pool
48	VP-04	Vernal Pool
49	VP-05	Vernal Pool
50	VP-06	Vernal Pool
51	VP-07	Vernal Pool
52	VP-08	Vernal Pool

```
veg |>
  filter(str_detect(transect_name, "GL|PU")) |>
  distinct(transect_name, habitat, psoc, cover_category) |>
  arrange(transect_name, cover_category) |>
  print(n = Inf)
```

A tibble: 701 x 4

	transect_name	habitat	psoc	cover_category
	<chr>	<chr>	<chr>	<chr>
1	GL-02	Perennial Grassland	Bare Ground	BARE GROUND
2	GL-02	Perennial Grassland	Epilobium_brachycarpum	NATIVE COVER
3	GL-02	Perennial Grassland	Erigeron_canadensis	NATIVE COVER
4	GL-02	Perennial Grassland	Heterotheca_grandiflora	NATIVE COVER
5	GL-02	Perennial Grassland	Stipa_pulchra	NATIVE COVER
6	GL-02	Perennial Grassland	Symphyotrichum_subulatum	NATIVE COVER
7	GL-02	Perennial Grassland	Baccharis_pilularis	NATIVE COVER
8	GL-02	Perennial Grassland	Juncus_bufonius	NATIVE COVER
9	GL-02	Perennial Grassland	Deinandra_fasciculata	NATIVE COVER
10	GL-02	Perennial Grassland	Eleocharis_acicularis	NATIVE COVER
11	GL-02	Perennial Grassland	Hordeum_brachyantherum_bra~	NATIVE COVER
12	GL-02	Perennial Grassland	Epilobium_ciliatum	NATIVE COVER
13	GL-02	Perennial Grassland	Festuca_microstachys	NATIVE COVER
14	GL-02	Perennial Grassland	Bromus_hordeaceus	NON-NATIVE CO~
15	GL-02	Perennial Grassland	Festuca_perennis	NON-NATIVE CO~
16	GL-02	Perennial Grassland	Helminthotheca_echioides	NON-NATIVE CO~
17	GL-02	Perennial Grassland	Medicago_polymorpha	NON-NATIVE CO~
18	GL-02	Perennial Grassland	Malva_parviflora	NON-NATIVE CO~
19	GL-02	Perennial Grassland	Polypogon_monspeliensis	NON-NATIVE CO~
20	GL-02	Perennial Grassland	Sonchus_oleraceus	NON-NATIVE CO~

21	GL-02	Perennial Grassland	Brachypodium_distachyon	NON-NATIVE CO~
22	GL-02	Perennial Grassland	Medicago_lupulina	NON-NATIVE CO~
23	GL-02	Perennial Grassland	Pseudognaphalium_luteoalbum	NON-NATIVE CO~
24	GL-02	Perennial Grassland	Melilotus_indicus	NON-NATIVE CO~
25	GL-02	Perennial Grassland	Festuca_myuros	NON-NATIVE CO~
26	GL-02	Perennial Grassland	Spergularia_sp	NON-NATIVE CO~
27	GL-02	Perennial Grassland	Avena_barbata	NON-NATIVE CO~
28	GL-02	Perennial Grassland	Hordeum_marinum	NON-NATIVE CO~
29	GL-02	Perennial Grassland	Logfia_gallica	NON-NATIVE CO~
30	GL-02	Perennial Grassland	Sonchus_sp	NON-NATIVE CO~
31	GL-02	Perennial Grassland	Avena_fatua	NON-NATIVE CO~
32	GL-02	Perennial Grassland	Lysimachia_arvensis	NON-NATIVE CO~
33	GL-02	Perennial Grassland	Parapholis_incurva	NON-NATIVE CO~
34	GL-02	Perennial Grassland	Plantago_coronopus	NON-NATIVE CO~
35	GL-02	Perennial Grassland	Trifolium_hirtum	NON-NATIVE CO~
36	GL-02	Perennial Grassland	Bromus_diandrus	NON-NATIVE CO~
37	GL-02	Perennial Grassland	Plantago_lanceolata	NON-NATIVE CO~
38	GL-02	Perennial Grassland	Sonchus_asper	NON-NATIVE CO~
39	GL-02	Perennial Grassland	Festuca_bromoides	NON-NATIVE CO~
40	GL-02	Perennial Grassland	Hordeum_murinum	NON-NATIVE CO~
41	GL-02	Perennial Grassland	unlisted	NON-NATIVE CO~
42	GL-02	Perennial Grassland	Dittrichia_graveolens	NON-NATIVE CO~
43	GL-02	Perennial Grassland	Centaurium_tenuiflorum	NON-NATIVE CO~
44	GL-02	Perennial Grassland	Lactuca_serriola	NON-NATIVE CO~
45	GL-02	Perennial Grassland	unlisted	OTHER COVER
46	GL-02	Perennial Grassland	Natural Thatch	THATCH
47	GL-07	Perennial Grassland	Bare Ground	BARE GROUND
48	GL-07	Perennial Grassland	Distichlis_spicata	NATIVE COVER
49	GL-07	Perennial Grassland	Stipa_pulchra	NATIVE COVER
50	GL-07	Perennial Grassland	Cressa_truxillensis	NATIVE COVER
51	GL-07	Perennial Grassland	Epilobium_brachycarpum	NATIVE COVER
52	GL-07	Perennial Grassland	Erigeron_canadensis	NATIVE COVER
53	GL-07	Perennial Grassland	Symphyotrichum_subulatum	NATIVE COVER
54	GL-07	Perennial Grassland	Stebbinsoseris_heterocarpa	NATIVE COVER
55	GL-07	Perennial Grassland	Hordeum_brachyantherum_bra~	NATIVE COVER
56	GL-07	Perennial Grassland	Deinandra_fasciculata	NATIVE COVER
57	GL-07	Perennial Grassland	Heterotheca_grandiflora	NATIVE COVER
58	GL-07	Perennial Grassland	Spergularia_macrotheca	NATIVE COVER
59	GL-07	Perennial Grassland	Baccharis_pilularis	NATIVE COVER
60	GL-07	Perennial Grassland	Avena_fatua	NON-NATIVE CO~
61	GL-07	Perennial Grassland	Festuca_perennis	NON-NATIVE CO~
62	GL-07	Perennial Grassland	Hordeum_marinum	NON-NATIVE CO~
63	GL-07	Perennial Grassland	Plantago_coronopus	NON-NATIVE CO~

64	GL-07	Perennial Grassland	Polypogon_monspeliensis	NON-NATIVE CO~
65	GL-07	Perennial Grassland	Bromus_catharticus	NON-NATIVE CO~
66	GL-07	Perennial Grassland	Bromus_hordeaceus	NON-NATIVE CO~
67	GL-07	Perennial Grassland	Medicago_polymorpha	NON-NATIVE CO~
68	GL-07	Perennial Grassland	Melilotus_indicus	NON-NATIVE CO~
69	GL-07	Perennial Grassland	Parapholis_incurva	NON-NATIVE CO~
70	GL-07	Perennial Grassland	Sonchus_sp	NON-NATIVE CO~
71	GL-07	Perennial Grassland	Brachypodium_distachyon	NON-NATIVE CO~
72	GL-07	Perennial Grassland	Avena_barbata	NON-NATIVE CO~
73	GL-07	Perennial Grassland	Hordeum_murinum	NON-NATIVE CO~
74	GL-07	Perennial Grassland	Cynodon_dactylon	NON-NATIVE CO~
75	GL-07	Perennial Grassland	Sonchus_asper	NON-NATIVE CO~
76	GL-07	Perennial Grassland	Bromus_diandrus	NON-NATIVE CO~
77	GL-07	Perennial Grassland	Malva_parviflora	NON-NATIVE CO~
78	GL-07	Perennial Grassland	Vicia_villosa	NON-NATIVE CO~
79	GL-07	Perennial Grassland	Centaurium_tenuiflorum	NON-NATIVE CO~
80	GL-07	Perennial Grassland	Festuca_myuros	NON-NATIVE CO~
81	GL-07	Perennial Grassland	Lysimachia_arvensis	NON-NATIVE CO~
82	GL-07	Perennial Grassland	Geranium_dissectum	NON-NATIVE CO~
83	GL-07	Perennial Grassland	Sonchus_oleraceus	NON-NATIVE CO~
84	GL-07	Perennial Grassland	Helminthotheca_echioides	NON-NATIVE CO~
85	GL-07	Perennial Grassland	Natural Thatch	THATCH
86	GL-09	Perennial Grassland	Bare Ground	BARE GROUND
87	GL-09	Perennial Grassland	Epilobium_brachycarpum	NATIVE COVER
88	GL-09	Perennial Grassland	Erigeron_canadensis	NATIVE COVER
89	GL-09	Perennial Grassland	Stipa_pulchra	NATIVE COVER
90	GL-09	Perennial Grassland	Symphyotrichum_subulatum	NATIVE COVER
91	GL-09	Perennial Grassland	Distichlis_spicata	NATIVE COVER
92	GL-09	Perennial Grassland	Heterotheca_grandiflora	NATIVE COVER
93	GL-09	Perennial Grassland	Lepidium_nitidum	NATIVE COVER
94	GL-09	Perennial Grassland	Plantago_erecta	NATIVE COVER
95	GL-09	Perennial Grassland	Sisyrinchium_bellum	NATIVE COVER
96	GL-09	Perennial Grassland	Stebbinsoseris_heterocarpa	NATIVE COVER
97	GL-09	Perennial Grassland	Deinandra_fasciculata	NATIVE COVER
98	GL-09	Perennial Grassland	Hordeum_brachyantherum_bra~	NATIVE COVER
99	GL-09	Perennial Grassland	Baccharis_pilularis	NATIVE COVER
100	GL-09	Perennial Grassland	Epilobium_ciliatum	NATIVE COVER
101	GL-09	Perennial Grassland	Laennecia_coulteri	NATIVE COVER
102	GL-09	Perennial Grassland	Spergularia_marina	NATIVE COVER
103	GL-09	Perennial Grassland	Festuca_microstachys	NATIVE COVER
104	GL-09	Perennial Grassland	Brachypodium_distachyon	NON-NATIVE CO~
105	GL-09	Perennial Grassland	Erigeron_bonariensis	NON-NATIVE CO~
106	GL-09	Perennial Grassland	Festuca_myuros	NON-NATIVE CO~

107	GL-09	Perennial Grassland	Festuca_perennis	NON-NATIVE CO~
108	GL-09	Perennial Grassland	Helminthotheca_echioides	NON-NATIVE CO~
109	GL-09	Perennial Grassland	Lythrum_hyssopifolia	NON-NATIVE CO~
110	GL-09	Perennial Grassland	Melilotus_indicus	NON-NATIVE CO~
111	GL-09	Perennial Grassland	Polypogon_monspeliensis	NON-NATIVE CO~
112	GL-09	Perennial Grassland	Bromus_diandrus	NON-NATIVE CO~
113	GL-09	Perennial Grassland	Bromus_hordeaceus	NON-NATIVE CO~
114	GL-09	Perennial Grassland	Centaureum_sp.	NON-NATIVE CO~
115	GL-09	Perennial Grassland	Geranium_dissectum	NON-NATIVE CO~
116	GL-09	Perennial Grassland	Hordeum_murinum	NON-NATIVE CO~
117	GL-09	Perennial Grassland	Medicago_polymorpha	NON-NATIVE CO~
118	GL-09	Perennial Grassland	Plantago_lanceolata	NON-NATIVE CO~
119	GL-09	Perennial Grassland	Sonchus_oleraceus	NON-NATIVE CO~
120	GL-09	Perennial Grassland	Brassica_nigra	NON-NATIVE CO~
121	GL-09	Perennial Grassland	Sonchus_sp	NON-NATIVE CO~
122	GL-09	Perennial Grassland	Lysimachia_arvensis	NON-NATIVE CO~
123	GL-09	Perennial Grassland	Avena_barbata	NON-NATIVE CO~
124	GL-09	Perennial Grassland	Spergularia_sp	NON-NATIVE CO~
125	GL-09	Perennial Grassland	Avena_fatua	NON-NATIVE CO~
126	GL-09	Perennial Grassland	Hordeum_marinum	NON-NATIVE CO~
127	GL-09	Perennial Grassland	Centaureum_tenuiflorum	NON-NATIVE CO~
128	GL-09	Perennial Grassland	Vicia_villosa	NON-NATIVE CO~
129	GL-09	Perennial Grassland	Vicia_sativa	NON-NATIVE CO~
130	GL-09	Perennial Grassland	Pseudognaphalium_luteoalbum	NON-NATIVE CO~
131	GL-09	Perennial Grassland	Festuca_bromoides	NON-NATIVE CO~
132	GL-09	Perennial Grassland	Sonchus_asper	NON-NATIVE CO~
133	GL-09	Perennial Grassland	Plantago_coronopus	NON-NATIVE CO~
134	GL-09	Perennial Grassland	unlisted	OTHER COVER
135	GL-09	Perennial Grassland	Natural Thatch	THATCH
136	GL-10	Perennial Grassland	Bare Ground	BARE GROUND
137	GL-10	Perennial Grassland	Baccharis_pilularis	NATIVE COVER
138	GL-10	Perennial Grassland	Erigeron_canadensis	NATIVE COVER
139	GL-10	Perennial Grassland	Stipa_pulchra	NATIVE COVER
140	GL-10	Perennial Grassland	Distichlis_spicata	NATIVE COVER
141	GL-10	Perennial Grassland	Epilobium_brachycarpum	NATIVE COVER
142	GL-10	Perennial Grassland	Symphyotrichum_subulatum	NATIVE COVER
143	GL-10	Perennial Grassland	Heterotheca_grandiflora	NATIVE COVER
144	GL-10	Perennial Grassland	Stebbinsoseris_heterocarpa	NATIVE COVER
145	GL-10	Perennial Grassland	Madia_sativa	NATIVE COVER
146	GL-10	Perennial Grassland	Deinandra_fasciculata	NATIVE COVER
147	GL-10	Perennial Grassland	unlisted	NATIVE COVER
148	GL-10	Perennial Grassland	Bromus_hordeaceus	NON-NATIVE CO~
149	GL-10	Perennial Grassland	Plantago_lanceolata	NON-NATIVE CO~

150	GL-10	Perennial	Grassland	Brachypodium_distachyon	NON-NATIVE CO~
151	GL-10	Perennial	Grassland	Bromus_diandrus	NON-NATIVE CO~
152	GL-10	Perennial	Grassland	Carduus_pycnocephalus	NON-NATIVE CO~
153	GL-10	Perennial	Grassland	Festuca_perennis	NON-NATIVE CO~
154	GL-10	Perennial	Grassland	Helminthotheca_echioides	NON-NATIVE CO~
155	GL-10	Perennial	Grassland	Medicago_polymorpha	NON-NATIVE CO~
156	GL-10	Perennial	Grassland	Sonchus_sp	NON-NATIVE CO~
157	GL-10	Perennial	Grassland	Polypogon_monspeliensis	NON-NATIVE CO~
158	GL-10	Perennial	Grassland	Melilotus_indicus	NON-NATIVE CO~
159	GL-10	Perennial	Grassland	Bromus_catharticus	NON-NATIVE CO~
160	GL-10	Perennial	Grassland	Brassica_nigra	NON-NATIVE CO~
161	GL-10	Perennial	Grassland	Erodium_cicutarium	NON-NATIVE CO~
162	GL-10	Perennial	Grassland	Lactuca_serriola	NON-NATIVE CO~
163	GL-10	Perennial	Grassland	Vicia_villosa	NON-NATIVE CO~
164	GL-10	Perennial	Grassland	Lysimachia_arvensis	NON-NATIVE CO~
165	GL-10	Perennial	Grassland	Centaurium_tenuiflorum	NON-NATIVE CO~
166	GL-10	Perennial	Grassland	Sonchus_oleraceus	NON-NATIVE CO~
167	GL-10	Perennial	Grassland	Pseudognaphalium_luteoalbum	NON-NATIVE CO~
168	GL-10	Perennial	Grassland	Festuca_myuros	NON-NATIVE CO~
169	GL-10	Perennial	Grassland	Trifolium_hirtum	NON-NATIVE CO~
170	GL-10	Perennial	Grassland	Natural Thatch	THATCH
171	GL-11	Perennial	Grassland	Bare Ground	BARE GROUND
172	GL-11	Perennial	Grassland	Stipa_pulchra	NATIVE COVER
173	GL-11	Perennial	Grassland	Erigeron_canadensis	NATIVE COVER
174	GL-11	Perennial	Grassland	Heterotheca_grandiflora	NATIVE COVER
175	GL-11	Perennial	Grassland	Deinandra_fasciculata	NATIVE COVER
176	GL-11	Perennial	Grassland	Epilobium_brachycarpum	NATIVE COVER
177	GL-11	Perennial	Grassland	Juncus_bufonius	NATIVE COVER
178	GL-11	Perennial	Grassland	Symphyotrichum_subulatum	NATIVE COVER
179	GL-11	Perennial	Grassland	Ambrosia_psilostachya	NATIVE COVER
180	GL-11	Perennial	Grassland	Hordeum_brachyantherum_bra~	NATIVE COVER
181	GL-11	Perennial	Grassland	Avena_fatua	NON-NATIVE CO~
182	GL-11	Perennial	Grassland	Brachypodium_distachyon	NON-NATIVE CO~
183	GL-11	Perennial	Grassland	Bromus_hordeaceus	NON-NATIVE CO~
184	GL-11	Perennial	Grassland	Festuca_myuros	NON-NATIVE CO~
185	GL-11	Perennial	Grassland	Hordeum_murinum	NON-NATIVE CO~
186	GL-11	Perennial	Grassland	Plantago_lanceolata	NON-NATIVE CO~
187	GL-11	Perennial	Grassland	Bromus_diandrus	NON-NATIVE CO~
188	GL-11	Perennial	Grassland	Festuca_perennis	NON-NATIVE CO~
189	GL-11	Perennial	Grassland	Plantago_coronopus	NON-NATIVE CO~
190	GL-11	Perennial	Grassland	Avena_barbata	NON-NATIVE CO~
191	GL-11	Perennial	Grassland	Sonchus_asper	NON-NATIVE CO~
192	GL-11	Perennial	Grassland	Logfia_gallica	NON-NATIVE CO~

193	GL-11	Perennial	Grassland	Parapholis_incurva	NON-NATIVE CO~
194	GL-11	Perennial	Grassland	Centaurium_tenuiflorum	NON-NATIVE CO~
195	GL-11	Perennial	Grassland	Melilotus_indicus	NON-NATIVE CO~
196	GL-11	Perennial	Grassland	Trifolium_hirtum	NON-NATIVE CO~
197	GL-11	Perennial	Grassland	Lysimachia_arvensis	NON-NATIVE CO~
198	GL-11	Perennial	Grassland	Vicia_villosa	NON-NATIVE CO~
199	GL-11	Perennial	Grassland	Geranium_dissectum	NON-NATIVE CO~
200	GL-11	Perennial	Grassland	Medicago_polymorpha	NON-NATIVE CO~
201	GL-11	Perennial	Grassland	Polypogon_monspeliensis	NON-NATIVE CO~
202	GL-11	Perennial	Grassland	Festuca_bromoides	NON-NATIVE CO~
203	GL-11	Perennial	Grassland	Helminthotheca_echioides	NON-NATIVE CO~
204	GL-11	Perennial	Grassland	Hordeum_marinum	NON-NATIVE CO~
205	GL-11	Perennial	Grassland	Sonchus_oleraceus	NON-NATIVE CO~
206	GL-11	Perennial	Grassland	Vicia_sativa	NON-NATIVE CO~
207	GL-11	Perennial	Grassland	Bromus_madritensis	NON-NATIVE CO~
208	GL-11	Perennial	Grassland	Natural Thatch	THATCH
209	GL-12	Perennial	Grassland	Bare Ground	BARE GROUND
210	GL-12	Perennial	Grassland	Baccharis_pilularis	NATIVE COVER
211	GL-12	Perennial	Grassland	Heterotheca_grandiflora	NATIVE COVER
212	GL-12	Perennial	Grassland	Hordeum_brachyantherum_bra~	NATIVE COVER
213	GL-12	Perennial	Grassland	Stipa_pulchra	NATIVE COVER
214	GL-12	Perennial	Grassland	Erigeron_canadensis	NATIVE COVER
215	GL-12	Perennial	Grassland	Sisyrinchium_bellum	NATIVE COVER
216	GL-12	Perennial	Grassland	Epilobium_brachycarpum	NATIVE COVER
217	GL-12	Perennial	Grassland	Sidalcea_malviflora	NATIVE COVER
218	GL-12	Perennial	Grassland	Lepidium_nitidum	NATIVE COVER
219	GL-12	Perennial	Grassland	Symphyotrichum_subulatum	NATIVE COVER
220	GL-12	Perennial	Grassland	Plantago_erecta	NATIVE COVER
221	GL-12	Perennial	Grassland	Stebbinsoseris_heterocarpa	NATIVE COVER
222	GL-12	Perennial	Grassland	Deinandra_fasciculata	NATIVE COVER
223	GL-12	Perennial	Grassland	Madia_sativa	NATIVE COVER
224	GL-12	Perennial	Grassland	Acmispon_maritimus	NATIVE COVER
225	GL-12	Perennial	Grassland	Spergularia_macrotheca	NATIVE COVER
226	GL-12	Perennial	Grassland	Centromadia_parryi_austral~	NATIVE COVER
227	GL-12	Perennial	Grassland	Brachypodium_distachyon	NON-NATIVE CO~
228	GL-12	Perennial	Grassland	Bromus_diandrus	NON-NATIVE CO~
229	GL-12	Perennial	Grassland	Festuca_perennis	NON-NATIVE CO~
230	GL-12	Perennial	Grassland	Geranium_dissectum	NON-NATIVE CO~
231	GL-12	Perennial	Grassland	Helminthotheca_echioides	NON-NATIVE CO~
232	GL-12	Perennial	Grassland	Lysimachia_arvensis	NON-NATIVE CO~
233	GL-12	Perennial	Grassland	Plantago_lanceolata	NON-NATIVE CO~
234	GL-12	Perennial	Grassland	Bromus_hordeaceus	NON-NATIVE CO~
235	GL-12	Perennial	Grassland	Avena_fatua	NON-NATIVE CO~

236	GL-12	Perennial Grassland	Plantago_coronopus	NON-NATIVE CO~
237	GL-12	Perennial Grassland	Hypochaeris_glabra	NON-NATIVE CO~
238	GL-12	Perennial Grassland	Sonchus_sp	NON-NATIVE CO~
239	GL-12	Perennial Grassland	Trifolium_hirtum	NON-NATIVE CO~
240	GL-12	Perennial Grassland	unlisted	NON-NATIVE CO~
241	GL-12	Perennial Grassland	Polypogon_monspeliensis	NON-NATIVE CO~
242	GL-12	Perennial Grassland	Medicago_polymorpha	NON-NATIVE CO~
243	GL-12	Perennial Grassland	Melilotus_albus	NON-NATIVE CO~
244	GL-12	Perennial Grassland	Parapholis_incurva	NON-NATIVE CO~
245	GL-12	Perennial Grassland	Hordeum_murinum	NON-NATIVE CO~
246	GL-12	Perennial Grassland	Avena_barbata	NON-NATIVE CO~
247	GL-12	Perennial Grassland	Erodium_cicutarium	NON-NATIVE CO~
248	GL-12	Perennial Grassland	Poa_annua	NON-NATIVE CO~
249	GL-12	Perennial Grassland	Sonchus_asper	NON-NATIVE CO~
250	GL-12	Perennial Grassland	Sonchus_oleraceus	NON-NATIVE CO~
251	GL-12	Perennial Grassland	Centaurium_tenuiflorum	NON-NATIVE CO~
252	GL-12	Perennial Grassland	Festuca_myuros	NON-NATIVE CO~
253	GL-12	Perennial Grassland	Erigeron_bonariensis	NON-NATIVE CO~
254	GL-12	Perennial Grassland	Melilotus_indicus	NON-NATIVE CO~
255	GL-12	Perennial Grassland	Lactuca_serriola	NON-NATIVE CO~
256	GL-12	Perennial Grassland	Natural Thatch	THATCH
257	GL-14	Perennial Grassland	Bare Ground	BARE GROUND
258	GL-14	Perennial Grassland	Epilobium_brachycarpum	NATIVE COVER
259	GL-14	Perennial Grassland	Erigeron_canadensis	NATIVE COVER
260	GL-14	Perennial Grassland	Juncus_phaeocephalus	NATIVE COVER
261	GL-14	Perennial Grassland	Stipa_pulchra	NATIVE COVER
262	GL-14	Perennial Grassland	Symphyotrichum_subulatum	NATIVE COVER
263	GL-14	Perennial Grassland	Juncus_bufonius	NATIVE COVER
264	GL-14	Perennial Grassland	Sisyrinchium_bellum	NATIVE COVER
265	GL-14	Perennial Grassland	Eleocharis_macrostachya	NATIVE COVER
266	GL-14	Perennial Grassland	Eryngium_vaseyi	NATIVE COVER
267	GL-14	Perennial Grassland	Deinandra_fasciculata	NATIVE COVER
268	GL-14	Perennial Grassland	Psilocarphus_tennulus	NATIVE COVER
269	GL-14	Perennial Grassland	Plantago_erecta	NATIVE COVER
270	GL-14	Perennial Grassland	Hordeum_brachyantherum_bra~	NATIVE COVER
271	GL-14	Perennial Grassland	Eleocharis_palustris	NATIVE COVER
272	GL-14	Perennial Grassland	Psilocarphus_brevissimus	NATIVE COVER
273	GL-14	Perennial Grassland	Bromus_hordeaceus	NON-NATIVE CO~
274	GL-14	Perennial Grassland	Festuca_myuros	NON-NATIVE CO~
275	GL-14	Perennial Grassland	Festuca_perennis	NON-NATIVE CO~
276	GL-14	Perennial Grassland	Lysimachia_arvensis	NON-NATIVE CO~
277	GL-14	Perennial Grassland	Medicago_polymorpha	NON-NATIVE CO~
278	GL-14	Perennial Grassland	Plantago_lanceolata	NON-NATIVE CO~

279	GL-14	Perennial Grassland	Polypogon_monspeliensis	NON-NATIVE CO~
280	GL-14	Perennial Grassland	Sonchus_sp	NON-NATIVE CO~
281	GL-14	Perennial Grassland	Melilotus_indicus	NON-NATIVE CO~
282	GL-14	Perennial Grassland	Plantago_coronopus	NON-NATIVE CO~
283	GL-14	Perennial Grassland	Centaurium_sp.	NON-NATIVE CO~
284	GL-14	Perennial Grassland	Geranium_dissectum	NON-NATIVE CO~
285	GL-14	Perennial Grassland	Helminthotheca_echioides	NON-NATIVE CO~
286	GL-14	Perennial Grassland	unlisted	NON-NATIVE CO~
287	GL-14	Perennial Grassland	Medicago_lupulina	NON-NATIVE CO~
288	GL-14	Perennial Grassland	Polypogon_viridis	NON-NATIVE CO~
289	GL-14	Perennial Grassland	Brachypodium_distachyon	NON-NATIVE CO~
290	GL-14	Perennial Grassland	Pseudognaphalium_luteoalbum	NON-NATIVE CO~
291	GL-14	Perennial Grassland	Avena_barbata	NON-NATIVE CO~
292	GL-14	Perennial Grassland	Sonchus_asper	NON-NATIVE CO~
293	GL-14	Perennial Grassland	Festuca_bromoides	NON-NATIVE CO~
294	GL-14	Perennial Grassland	Hypochaeris_glabra	NON-NATIVE CO~
295	GL-14	Perennial Grassland	Bromus_diandrus	NON-NATIVE CO~
296	GL-14	Perennial Grassland	Centaurium_tenuiflorum	NON-NATIVE CO~
297	GL-14	Perennial Grassland	Trifolium_hirtum	NON-NATIVE CO~
298	GL-14	Perennial Grassland	Lythrum_hyssopifolia	NON-NATIVE CO~
299	GL-14	Perennial Grassland	Crypsis_schoenoides	NON-NATIVE CO~
300	GL-14	Perennial Grassland	Polygonum_aviculare_depres~	NON-NATIVE CO~
301	GL-14	Perennial Grassland	Polypogon_interruptus	NON-NATIVE CO~
302	GL-14	Perennial Grassland	Trifolium_tomentosum	NON-NATIVE CO~
303	GL-14	Perennial Grassland	Vicia_villosa	NON-NATIVE CO~
304	GL-14	Perennial Grassland	unlisted	OTHER COVER
305	GL-14	Perennial Grassland	Natural Thatch	THATCH
306	GL-15	Perennial Grassland	Bare Ground	BARE GROUND
307	GL-15	Perennial Grassland	Erigeron_canadensis	NATIVE COVER
308	GL-15	Perennial Grassland	Heterotheca_grandiflora	NATIVE COVER
309	GL-15	Perennial Grassland	Stipa_pulchra	NATIVE COVER
310	GL-15	Perennial Grassland	Deinandra_fasciculata	NATIVE COVER
311	GL-15	Perennial Grassland	Symphyotrichum_subulatum	NATIVE COVER
312	GL-15	Perennial Grassland	Acmispon_maritimus	NATIVE COVER
313	GL-15	Perennial Grassland	Festuca_microstachys	NATIVE COVER
314	GL-15	Perennial Grassland	Acmispon_wrangelianus	NATIVE COVER
315	GL-15	Perennial Grassland	Plantago_erecta	NATIVE COVER
316	GL-15	Perennial Grassland	Juncus_bufonius	NATIVE COVER
317	GL-15	Perennial Grassland	Acmispon_glaber	NATIVE COVER
318	GL-15	Perennial Grassland	Stebbinsoseris_heterocarpa	NATIVE COVER
319	GL-15	Perennial Grassland	Epilobium_brachycarpum	NATIVE COVER
320	GL-15	Perennial Grassland	Spergularia_marina	NATIVE COVER
321	GL-15	Perennial Grassland	Hordeum_brachyantherum	NATIVE COVER

322	GL-15	Perennial Grassland	Elymus_triticoides	NATIVE COVER
323	GL-15	Perennial Grassland	Bromus_hordeaceus	NON-NATIVE CO~
324	GL-15	Perennial Grassland	Festuca_myuros	NON-NATIVE CO~
325	GL-15	Perennial Grassland	Festuca_perennis	NON-NATIVE CO~
326	GL-15	Perennial Grassland	Medicago_polymorpha	NON-NATIVE CO~
327	GL-15	Perennial Grassland	Melilotus_indicus	NON-NATIVE CO~
328	GL-15	Perennial Grassland	Polypogon_monspeliensis	NON-NATIVE CO~
329	GL-15	Perennial Grassland	Sonchus_sp	NON-NATIVE CO~
330	GL-15	Perennial Grassland	Avena_fatua	NON-NATIVE CO~
331	GL-15	Perennial Grassland	Brachypodium_distachyon	NON-NATIVE CO~
332	GL-15	Perennial Grassland	Hypochaeris_glabra	NON-NATIVE CO~
333	GL-15	Perennial Grassland	Plantago_coronopus	NON-NATIVE CO~
334	GL-15	Perennial Grassland	Cortaderia_selloana	NON-NATIVE CO~
335	GL-15	Perennial Grassland	Helminthotheca_echioides	NON-NATIVE CO~
336	GL-15	Perennial Grassland	Malva_parviflora	NON-NATIVE CO~
337	GL-15	Perennial Grassland	Plantago_lanceolata	NON-NATIVE CO~
338	GL-15	Perennial Grassland	Lysimachia_arvensis	NON-NATIVE CO~
339	GL-15	Perennial Grassland	Pseudognaphalium_luteoalbum	NON-NATIVE CO~
340	GL-15	Perennial Grassland	Bromus_diandrus	NON-NATIVE CO~
341	GL-15	Perennial Grassland	Erodium_cicutarium	NON-NATIVE CO~
342	GL-15	Perennial Grassland	Sonchus_asper	NON-NATIVE CO~
343	GL-15	Perennial Grassland	Hordeum_marinum	NON-NATIVE CO~
344	GL-15	Perennial Grassland	Logfia_gallica	NON-NATIVE CO~
345	GL-15	Perennial Grassland	Melilotus_albus	NON-NATIVE CO~
346	GL-15	Perennial Grassland	Centaurium_tenuiflorum	NON-NATIVE CO~
347	GL-15	Perennial Grassland	Sonchus_oleraceus	NON-NATIVE CO~
348	GL-15	Perennial Grassland	Festuca_bromoides	NON-NATIVE CO~
349	GL-15	Perennial Grassland	Polygonum_aviculare_depres~	NON-NATIVE CO~
350	GL-15	Perennial Grassland	unlisted	OTHER COVER
351	GL-15	Perennial Grassland	moss	OTHER COVER
352	GL-15	Perennial Grassland	boulder_rock	OTHER COVER
353	GL-15	Perennial Grassland	biocrust	OTHER COVER
354	GL-15	Perennial Grassland	Natural Thatch	THATCH
355	PU-01	Peripheral Uplands	Bare Ground	BARE GROUND
356	PU-01	Peripheral Uplands	Artemisia_californica	NATIVE COVER
357	PU-01	Peripheral Uplands	Deinandra_fasciculata	NATIVE COVER
358	PU-01	Peripheral Uplands	Elymus_condensatus	NATIVE COVER
359	PU-01	Peripheral Uplands	Isocoma_menziesii	NATIVE COVER
360	PU-01	Peripheral Uplands	Baccharis_pilularis	NATIVE COVER
361	PU-01	Peripheral Uplands	Epilobium_brachycarpum	NATIVE COVER
362	PU-01	Peripheral Uplands	Bromus_diandrus	NON-NATIVE CO~
363	PU-01	Peripheral Uplands	Bromus_hordeaceus	NON-NATIVE CO~
364	PU-01	Peripheral Uplands	Festuca_myuros	NON-NATIVE CO~

365	PU-01	Peripheral Uplands	Festuca_perennis	NON-NATIVE CO~
366	PU-01	Peripheral Uplands	Melilotus_indicus	NON-NATIVE CO~
367	PU-01	Peripheral Uplands	Plantago_coronopus	NON-NATIVE CO~
368	PU-01	Peripheral Uplands	Spergularia_sp	NON-NATIVE CO~
369	PU-01	Peripheral Uplands	Hordeum_marinum	NON-NATIVE CO~
370	PU-01	Peripheral Uplands	Plantago_lanceolata	NON-NATIVE CO~
371	PU-01	Peripheral Uplands	Atriplex_semibaccata	NON-NATIVE CO~
372	PU-01	Peripheral Uplands	Bromus_madritensis	NON-NATIVE CO~
373	PU-01	Peripheral Uplands	Brachypodium_distachyon	NON-NATIVE CO~
374	PU-01	Peripheral Uplands	Medicago_polymorpha	NON-NATIVE CO~
375	PU-01	Peripheral Uplands	Vicia_sativa	NON-NATIVE CO~
376	PU-01	Peripheral Uplands	unlisted	OTHER COVER
377	PU-01	Peripheral Uplands	moss	OTHER COVER
378	PU-01	Peripheral Uplands	mulch/woodchips	OTHER COVER
379	PU-01	Peripheral Uplands	Natural Thatch	THATCH
380	PU-02	Peripheral Uplands	Bare Ground	BARE GROUND
381	PU-02	Peripheral Uplands	Distichlis_spicata	NATIVE COVER
382	PU-02	Peripheral Uplands	Epilobium_canum	NATIVE COVER
383	PU-02	Peripheral Uplands	Bromus_carinatus	NATIVE COVER
384	PU-02	Peripheral Uplands	Epilobium_brachycarpum	NATIVE COVER
385	PU-02	Peripheral Uplands	Quercus_agrifolia	NATIVE COVER
386	PU-02	Peripheral Uplands	Erigeron_canadensis	NATIVE COVER
387	PU-02	Peripheral Uplands	Eleocharis_macrostachya	NATIVE COVER
388	PU-02	Peripheral Uplands	Juncus_bufonius	NATIVE COVER
389	PU-02	Peripheral Uplands	Juncus_patens	NATIVE COVER
390	PU-02	Peripheral Uplands	Baccharis_pilularis	NATIVE COVER
391	PU-02	Peripheral Uplands	Spergula_marina	NATIVE COVER
392	PU-02	Peripheral Uplands	Symphotrichum_subulatum	NATIVE COVER
393	PU-02	Peripheral Uplands	Stipa_pulchra	NATIVE COVER
394	PU-02	Peripheral Uplands	Atriplex_lentiformis	NATIVE COVER
395	PU-02	Peripheral Uplands	Spergularia_marina	NATIVE COVER
396	PU-02	Peripheral Uplands	unlisted	NATIVE COVER
397	PU-02	Peripheral Uplands	Eleocharis_palustris	NATIVE COVER
398	PU-02	Peripheral Uplands	Brachypodium_distachyon	NON-NATIVE CO~
399	PU-02	Peripheral Uplands	Bromus_catharticus	NON-NATIVE CO~
400	PU-02	Peripheral Uplands	Festuca_perennis	NON-NATIVE CO~
401	PU-02	Peripheral Uplands	Helminthotheca_echioides	NON-NATIVE CO~
402	PU-02	Peripheral Uplands	Hordeum_marinum	NON-NATIVE CO~
403	PU-02	Peripheral Uplands	Malva_parviflora	NON-NATIVE CO~
404	PU-02	Peripheral Uplands	Plantago_coronopus	NON-NATIVE CO~
405	PU-02	Peripheral Uplands	Sonchus_sp	NON-NATIVE CO~
406	PU-02	Peripheral Uplands	Spergularia_sp	NON-NATIVE CO~
407	PU-02	Peripheral Uplands	Polygonum_aviculare_depres~	NON-NATIVE CO~

408	PU-02	Peripheral Uplands	Polypogon_monspeliensis	NON-NATIVE CO~
409	PU-02	Peripheral Uplands	Lythrum_hyssopifolia	NON-NATIVE CO~
410	PU-02	Peripheral Uplands	Polypogon_viridis	NON-NATIVE CO~
411	PU-02	Peripheral Uplands	Melilotus_indicus	NON-NATIVE CO~
412	PU-02	Peripheral Uplands	Plantago_lanceolata	NON-NATIVE CO~
413	PU-02	Peripheral Uplands	Avena_fatua	NON-NATIVE CO~
414	PU-02	Peripheral Uplands	Bromus_diandrus	NON-NATIVE CO~
415	PU-02	Peripheral Uplands	Festuca_myuros	NON-NATIVE CO~
416	PU-02	Peripheral Uplands	Sonchus_asper	NON-NATIVE CO~
417	PU-02	Peripheral Uplands	Cotula_coronopifolia	NON-NATIVE CO~
418	PU-02	Peripheral Uplands	Cynodon_dactylon	NON-NATIVE CO~
419	PU-02	Peripheral Uplands	Hordeum_murinum	NON-NATIVE CO~
420	PU-02	Peripheral Uplands	Centaurium_tenuiflorum	NON-NATIVE CO~
421	PU-02	Peripheral Uplands	Festuca_bromoides	NON-NATIVE CO~
422	PU-02	Peripheral Uplands	Lysimachia_arvensis	NON-NATIVE CO~
423	PU-02	Peripheral Uplands	Bromus_hordeaceus	NON-NATIVE CO~
424	PU-02	Peripheral Uplands	Medicago_polymorpha	NON-NATIVE CO~
425	PU-02	Peripheral Uplands	Poa_annua	NON-NATIVE CO~
426	PU-02	Peripheral Uplands	mulch/woodchips	OTHER COVER
427	PU-02	Peripheral Uplands	Natural Thatch	THATCH
428	PU-03	Peripheral Uplands	Bare Ground	BARE GROUND
429	PU-03	Peripheral Uplands	Distichlis_spicata	NATIVE COVER
430	PU-03	Peripheral Uplands	Epilobium_brachycarpum	NATIVE COVER
431	PU-03	Peripheral Uplands	Erigeron_canadensis	NATIVE COVER
432	PU-03	Peripheral Uplands	Sisyrinchium_bellum	NATIVE COVER
433	PU-03	Peripheral Uplands	Stipa_pulchra	NATIVE COVER
434	PU-03	Peripheral Uplands	Symphyotrichum_subulatum	NATIVE COVER
435	PU-03	Peripheral Uplands	Cyperus_eragrostis	NATIVE COVER
436	PU-03	Peripheral Uplands	Juncus_occidentalis	NATIVE COVER
437	PU-03	Peripheral Uplands	Baccharis_pilularis	NATIVE COVER
438	PU-03	Peripheral Uplands	Centromadia_parryi_austral~	NATIVE COVER
439	PU-03	Peripheral Uplands	Salix_lasiiolepis	NATIVE COVER
440	PU-03	Peripheral Uplands	Epilobium_ciliatum	NATIVE COVER
441	PU-03	Peripheral Uplands	Elymus_glaucus	NATIVE COVER
442	PU-03	Peripheral Uplands	Hordeum_brachyantherum	NATIVE COVER
443	PU-03	Peripheral Uplands	Bolboschoenus_maritimus	NATIVE COVER
444	PU-03	Peripheral Uplands	Juncus_phaeocephalus	NATIVE COVER
445	PU-03	Peripheral Uplands	Festuca_microstachys	NATIVE COVER
446	PU-03	Peripheral Uplands	Bromus_hordeaceus	NON-NATIVE CO~
447	PU-03	Peripheral Uplands	Erigeron_bonariensis	NON-NATIVE CO~
448	PU-03	Peripheral Uplands	Festuca_perennis	NON-NATIVE CO~
449	PU-03	Peripheral Uplands	Hordeum_murinum	NON-NATIVE CO~
450	PU-03	Peripheral Uplands	Lysimachia_arvensis	NON-NATIVE CO~

451	PU-03	Peripheral Uplands	Malva_parviflora	NON-NATIVE CO~
452	PU-03	Peripheral Uplands	Medicago_polymorpha	NON-NATIVE CO~
453	PU-03	Peripheral Uplands	Plantago_coronopus	NON-NATIVE CO~
454	PU-03	Peripheral Uplands	Sonchus_sp	NON-NATIVE CO~
455	PU-03	Peripheral Uplands	Cotula_coronopifolia	NON-NATIVE CO~
456	PU-03	Peripheral Uplands	Erodium_cicutarium	NON-NATIVE CO~
457	PU-03	Peripheral Uplands	Festuca_myuros	NON-NATIVE CO~
458	PU-03	Peripheral Uplands	Pseudognaphalium_luteoalbum	NON-NATIVE CO~
459	PU-03	Peripheral Uplands	Bromus_diandrus	NON-NATIVE CO~
460	PU-03	Peripheral Uplands	Bromus_catharticus	NON-NATIVE CO~
461	PU-03	Peripheral Uplands	Cynodon_dactylon	NON-NATIVE CO~
462	PU-03	Peripheral Uplands	Paspalum_dilatatum	NON-NATIVE CO~
463	PU-03	Peripheral Uplands	Pinus_pinea	NON-NATIVE CO~
464	PU-03	Peripheral Uplands	Polygonum_aviculare_depres~	NON-NATIVE CO~
465	PU-03	Peripheral Uplands	Vicia_sativa	NON-NATIVE CO~
466	PU-03	Peripheral Uplands	Centaurium_sp.	NON-NATIVE CO~
467	PU-03	Peripheral Uplands	Geranium_dissectum	NON-NATIVE CO~
468	PU-03	Peripheral Uplands	Polypogon_monspelienensis	NON-NATIVE CO~
469	PU-03	Peripheral Uplands	Lepidium_didymum	NON-NATIVE CO~
470	PU-03	Peripheral Uplands	Brachypodium_distachyon	NON-NATIVE CO~
471	PU-03	Peripheral Uplands	Helminthotheca_echioides	NON-NATIVE CO~
472	PU-03	Peripheral Uplands	Polypogon_viridis	NON-NATIVE CO~
473	PU-03	Peripheral Uplands	Atriplex_prostrata	NON-NATIVE CO~
474	PU-03	Peripheral Uplands	Erodium_botrys	NON-NATIVE CO~
475	PU-03	Peripheral Uplands	Melilotus_indicus	NON-NATIVE CO~
476	PU-03	Peripheral Uplands	Sonchus_oleraceus	NON-NATIVE CO~
477	PU-03	Peripheral Uplands	Lactuca_serriola	NON-NATIVE CO~
478	PU-03	Peripheral Uplands	Sonchus_asper	NON-NATIVE CO~
479	PU-03	Peripheral Uplands	Vicia_villosa	NON-NATIVE CO~
480	PU-03	Peripheral Uplands	Avena_fatua	NON-NATIVE CO~
481	PU-03	Peripheral Uplands	Stipa_miliacea	NON-NATIVE CO~
482	PU-03	Peripheral Uplands	Polypogon_interruptus	NON-NATIVE CO~
483	PU-03	Peripheral Uplands	Festuca_bromoides	NON-NATIVE CO~
484	PU-03	Peripheral Uplands	Rumex_crispus	NON-NATIVE CO~
485	PU-03	Peripheral Uplands	Hordeum_marinum	NON-NATIVE CO~
486	PU-03	Peripheral Uplands	Lythrum_hyssopifolia	NON-NATIVE CO~
487	PU-03	Peripheral Uplands	unlisted	OTHER COVER
488	PU-03	Peripheral Uplands	Natural Thatch	THATCH
489	PU-04	Peripheral Uplands	Bare Ground	BARE GROUND
490	PU-04	Peripheral Uplands	Distichlis_spicata	NATIVE COVER
491	PU-04	Peripheral Uplands	Epilobium_brachycarpum	NATIVE COVER
492	PU-04	Peripheral Uplands	Epilobium_canum	NATIVE COVER
493	PU-04	Peripheral Uplands	Malacothrix_saxatilis_v._t~	NATIVE COVER

494	PU-04	Peripheral Uplands	<i>Stipa_pulchra</i>	NATIVE COVER
495	PU-04	Peripheral Uplands	<i>Artemisia_californica</i>	NATIVE COVER
496	PU-04	Peripheral Uplands	<i>Encelia_californica</i>	NATIVE COVER
497	PU-04	Peripheral Uplands	<i>Eriophyllum_confertiflorum</i>	NATIVE COVER
498	PU-04	Peripheral Uplands	<i>Ambrosia_psilostachya</i>	NATIVE COVER
499	PU-04	Peripheral Uplands	<i>Elymus_triticoides</i>	NATIVE COVER
500	PU-04	Peripheral Uplands	<i>Heterotheca_grandiflora</i>	NATIVE COVER
501	PU-04	Peripheral Uplands	<i>Symphotrichum_subulatum</i>	NATIVE COVER
502	PU-04	Peripheral Uplands	<i>Erigeron_canadensis</i>	NATIVE COVER
503	PU-04	Peripheral Uplands	<i>Euphorbia_maculata</i>	NATIVE COVER
504	PU-04	Peripheral Uplands	<i>Malacothris_saxatilis_var_~</i>	NATIVE COVER
505	PU-04	Peripheral Uplands	<i>Baccharis_pilularis</i>	NATIVE COVER
506	PU-04	Peripheral Uplands	<i>Bromus_carinatus</i>	NATIVE COVER
507	PU-04	Peripheral Uplands	<i>Lupinus_succulentus</i>	NATIVE COVER
508	PU-04	Peripheral Uplands	<i>Cressa_truxillensis</i>	NATIVE COVER
509	PU-04	Peripheral Uplands	<i>Datura_wrightii</i>	NATIVE COVER
510	PU-04	Peripheral Uplands	<i>Melica_imperfecta</i>	NATIVE COVER
511	PU-04	Peripheral Uplands	<i>Cyperus_eragrostis</i>	NATIVE COVER
512	PU-04	Peripheral Uplands	<i>Hordeum_brachyantherum_bra~</i>	NATIVE COVER
513	PU-04	Peripheral Uplands	<i>Extriplex_californica</i>	NATIVE COVER
514	PU-04	Peripheral Uplands	<i>Quercus_agrifolia</i>	NATIVE COVER
515	PU-04	Peripheral Uplands	<i>Bromus_hordeaceus</i>	NON-NATIVE CO~
516	PU-04	Peripheral Uplands	<i>Festuca_myuros</i>	NON-NATIVE CO~
517	PU-04	Peripheral Uplands	<i>Festuca_perennis</i>	NON-NATIVE CO~
518	PU-04	Peripheral Uplands	<i>Geranium_dissectum</i>	NON-NATIVE CO~
519	PU-04	Peripheral Uplands	<i>Sonchus_oleraceus</i>	NON-NATIVE CO~
520	PU-04	Peripheral Uplands	<i>Bromus_diandrus</i>	NON-NATIVE CO~
521	PU-04	Peripheral Uplands	<i>Medicago_polymorpha</i>	NON-NATIVE CO~
522	PU-04	Peripheral Uplands	<i>Polypogon_interruptus</i>	NON-NATIVE CO~
523	PU-04	Peripheral Uplands	<i>Pseudognaphalium_luteoalbum</i>	NON-NATIVE CO~
524	PU-04	Peripheral Uplands	<i>Erigeron_bonariensis</i>	NON-NATIVE CO~
525	PU-04	Peripheral Uplands	<i>Helminthotheca_echioides</i>	NON-NATIVE CO~
526	PU-04	Peripheral Uplands	<i>Lactuca_serriola</i>	NON-NATIVE CO~
527	PU-04	Peripheral Uplands	<i>Polypogon_monspeliensis</i>	NON-NATIVE CO~
528	PU-04	Peripheral Uplands	<i>Trifolium_hirtum</i>	NON-NATIVE CO~
529	PU-04	Peripheral Uplands	<i>Plantago_lanceolata</i>	NON-NATIVE CO~
530	PU-04	Peripheral Uplands	<i>Bromus_catharticus</i>	NON-NATIVE CO~
531	PU-04	Peripheral Uplands	<i>Lysimachia_arvensis</i>	NON-NATIVE CO~
532	PU-04	Peripheral Uplands	<i>Stipa_miliacea</i>	NON-NATIVE CO~
533	PU-04	Peripheral Uplands	<i>Lepidium_didymum</i>	NON-NATIVE CO~
534	PU-04	Peripheral Uplands	<i>Malva_parviflora</i>	NON-NATIVE CO~
535	PU-04	Peripheral Uplands	<i>Polygonum_aviculare_depres~</i>	NON-NATIVE CO~
536	PU-04	Peripheral Uplands	<i>Polypogon_viridis</i>	NON-NATIVE CO~

537	PU-04	Peripheral Uplands	Sonchus_asper	NON-NATIVE CO~
538	PU-04	Peripheral Uplands	Medicago_lupulina	NON-NATIVE CO~
539	PU-04	Peripheral Uplands	Avena_fatua	NON-NATIVE CO~
540	PU-04	Peripheral Uplands	Avena_barbata	NON-NATIVE CO~
541	PU-04	Peripheral Uplands	Hordeum_murinum	NON-NATIVE CO~
542	PU-04	Peripheral Uplands	Vicia_villosa	NON-NATIVE CO~
543	PU-04	Peripheral Uplands	Paspalum_dilatatum	NON-NATIVE CO~
544	PU-04	Peripheral Uplands	Brachypodium_distachyon	NON-NATIVE CO~
545	PU-04	Peripheral Uplands	Erodium_cicutarium	NON-NATIVE CO~
546	PU-04	Peripheral Uplands	Centaurium_tenuiflorum	NON-NATIVE CO~
547	PU-04	Peripheral Uplands	Vicia_sativa	NON-NATIVE CO~
548	PU-04	Peripheral Uplands	Euphorbia_serpens	NON-NATIVE CO~
549	PU-04	Peripheral Uplands	Festuca_bromoides	NON-NATIVE CO~
550	PU-04	Peripheral Uplands	Erodium_botrys	NON-NATIVE CO~
551	PU-04	Peripheral Uplands	unlisted	NON-NATIVE CO~
552	PU-04	Peripheral Uplands	unlisted	OTHER COVER
553	PU-04	Peripheral Uplands	Natural Thatch	THATCH
554	PU-05	Peripheral Uplands	Bare Ground	BARE GROUND
555	PU-05	Peripheral Uplands	Baccharis_pilularis	NATIVE COVER
556	PU-05	Peripheral Uplands	Cyperus_eragrostis	NATIVE COVER
557	PU-05	Peripheral Uplands	Eleocharis_macrostachya	NATIVE COVER
558	PU-05	Peripheral Uplands	Erigeron_canadensis	NATIVE COVER
559	PU-05	Peripheral Uplands	Frankenia_salina	NATIVE COVER
560	PU-05	Peripheral Uplands	Symphotrichum_subulatum	NATIVE COVER
561	PU-05	Peripheral Uplands	Encelia_californica	NATIVE COVER
562	PU-05	Peripheral Uplands	Eriogonum_parvifolium	NATIVE COVER
563	PU-05	Peripheral Uplands	Lonicera_subspicata_v._sub~	NATIVE COVER
564	PU-05	Peripheral Uplands	Salvia_leucophylla	NATIVE COVER
565	PU-05	Peripheral Uplands	Eriophyllum_confertiflorum	NATIVE COVER
566	PU-05	Peripheral Uplands	Artemisia_californica	NATIVE COVER
567	PU-05	Peripheral Uplands	Distichlis_spicata	NATIVE COVER
568	PU-05	Peripheral Uplands	Laennecia_coulteri	NATIVE COVER
569	PU-05	Peripheral Uplands	Scrophularia_californica	NATIVE COVER
570	PU-05	Peripheral Uplands	Eleocharis_palustris	NATIVE COVER
571	PU-05	Peripheral Uplands	Epilobium_ciliatum	NATIVE COVER
572	PU-05	Peripheral Uplands	Heterotheca_grandiflora	NATIVE COVER
573	PU-05	Peripheral Uplands	Juncus_bufonius	NATIVE COVER
574	PU-05	Peripheral Uplands	Ambrosia_psilostachya	NATIVE COVER
575	PU-05	Peripheral Uplands	Festuca_myuros	NON-NATIVE CO~
576	PU-05	Peripheral Uplands	Helminthotheca_echioides	NON-NATIVE CO~
577	PU-05	Peripheral Uplands	Sonchus_oleraceus	NON-NATIVE CO~
578	PU-05	Peripheral Uplands	Avena_fatua	NON-NATIVE CO~
579	PU-05	Peripheral Uplands	Bromus_hordeaceus	NON-NATIVE CO~

580	PU-05	Peripheral Uplands	Convolvulus_arvensis	NON-NATIVE CO~
581	PU-05	Peripheral Uplands	Erigeron_bonariensis	NON-NATIVE CO~
582	PU-05	Peripheral Uplands	Festuca_perennis	NON-NATIVE CO~
583	PU-05	Peripheral Uplands	Bromus_diandrus	NON-NATIVE CO~
584	PU-05	Peripheral Uplands	Lysimachia_arvensis	NON-NATIVE CO~
585	PU-05	Peripheral Uplands	Parapholis_incurva	NON-NATIVE CO~
586	PU-05	Peripheral Uplands	Polypogon_interruptus	NON-NATIVE CO~
587	PU-05	Peripheral Uplands	Sonchus_asper	NON-NATIVE CO~
588	PU-05	Peripheral Uplands	Bromus_catharticus	NON-NATIVE CO~
589	PU-05	Peripheral Uplands	Polygonum_aviculare_depres~	NON-NATIVE CO~
590	PU-05	Peripheral Uplands	Stipa_miliacea	NON-NATIVE CO~
591	PU-05	Peripheral Uplands	Hordeum_marinum	NON-NATIVE CO~
592	PU-05	Peripheral Uplands	Hordeum_murinum	NON-NATIVE CO~
593	PU-05	Peripheral Uplands	Lepidium_didymum	NON-NATIVE CO~
594	PU-05	Peripheral Uplands	Avena_barbata	NON-NATIVE CO~
595	PU-05	Peripheral Uplands	Brachypodium_distachyon	NON-NATIVE CO~
596	PU-05	Peripheral Uplands	Euphorbia_serpens	NON-NATIVE CO~
597	PU-05	Peripheral Uplands	Festuca_bromoides	NON-NATIVE CO~
598	PU-05	Peripheral Uplands	Tragopogon_porrifolius	NON-NATIVE CO~
599	PU-05	Peripheral Uplands	Pennisetum_clandestinum	NON-NATIVE CO~
600	PU-05	Peripheral Uplands	Salsola_tragus	NON-NATIVE CO~
601	PU-05	Peripheral Uplands	Lythrum_hyssopifolia	NON-NATIVE CO~
602	PU-05	Peripheral Uplands	Polypogon_monspeliensis	NON-NATIVE CO~
603	PU-05	Peripheral Uplands	Centaurium_tenuiflorum	NON-NATIVE CO~
604	PU-05	Peripheral Uplands	Pseudognaphalium_luteoalbum	NON-NATIVE CO~
605	PU-05	Peripheral Uplands	Geranium_dissectum	NON-NATIVE CO~
606	PU-05	Peripheral Uplands	Medicago_polymorpha	NON-NATIVE CO~
607	PU-05	Peripheral Uplands	mulch/woodchips	OTHER COVER
608	PU-05	Peripheral Uplands	unlisted	OTHER COVER
609	PU-05	Peripheral Uplands	biocrust	OTHER COVER
610	PU-05	Peripheral Uplands	log	OTHER COVER
611	PU-05	Peripheral Uplands	Natural Thatch	THATCH
612	PU-06	Peripheral Uplands	Bare Ground	BARE GROUND
613	PU-06	Peripheral Uplands	Ambrosia_psilostachya	NATIVE COVER
614	PU-06	Peripheral Uplands	Cyperus_eragrostis	NATIVE COVER
615	PU-06	Peripheral Uplands	Erigeron_canadensis	NATIVE COVER
616	PU-06	Peripheral Uplands	Frankenia_salina	NATIVE COVER
617	PU-06	Peripheral Uplands	Distichlis_spicata	NATIVE COVER
618	PU-06	Peripheral Uplands	Elymus_triticoides	NATIVE COVER
619	PU-06	Peripheral Uplands	Scrophularia_californica	NATIVE COVER
620	PU-06	Peripheral Uplands	Artemisia_californica	NATIVE COVER
621	PU-06	Peripheral Uplands	Encelia_californica	NATIVE COVER
622	PU-06	Peripheral Uplands	Epilobium_brachycarpum	NATIVE COVER

623	PU-06	Peripheral Uplands	Eriogonum_parvifolium	NATIVE COVER
624	PU-06	Peripheral Uplands	Baccharis_pilularis	NATIVE COVER
625	PU-06	Peripheral Uplands	Eriophyllum_confertiflorum	NATIVE COVER
626	PU-06	Peripheral Uplands	Festuca_microstachys	NATIVE COVER
627	PU-06	Peripheral Uplands	Avena_barbata	NON-NATIVE CO~
628	PU-06	Peripheral Uplands	Bromus_diandrus	NON-NATIVE CO~
629	PU-06	Peripheral Uplands	Bromus_hordeaceus	NON-NATIVE CO~
630	PU-06	Peripheral Uplands	Cynodon_dactylon	NON-NATIVE CO~
631	PU-06	Peripheral Uplands	Festuca_perennis	NON-NATIVE CO~
632	PU-06	Peripheral Uplands	Helminthotheca_echioides	NON-NATIVE CO~
633	PU-06	Peripheral Uplands	Lysimachia_arvensis	NON-NATIVE CO~
634	PU-06	Peripheral Uplands	Medicago_polymorpha	NON-NATIVE CO~
635	PU-06	Peripheral Uplands	Melilotus_indicus	NON-NATIVE CO~
636	PU-06	Peripheral Uplands	Plantago_coronopus	NON-NATIVE CO~
637	PU-06	Peripheral Uplands	Plantago_lanceolata	NON-NATIVE CO~
638	PU-06	Peripheral Uplands	Rumex_crispus	NON-NATIVE CO~
639	PU-06	Peripheral Uplands	Sonchus_oleraceus	NON-NATIVE CO~
640	PU-06	Peripheral Uplands	Vicia_sativa	NON-NATIVE CO~
641	PU-06	Peripheral Uplands	Erigeron_bonariensis	NON-NATIVE CO~
642	PU-06	Peripheral Uplands	Polypogon_monspeliensis	NON-NATIVE CO~
643	PU-06	Peripheral Uplands	Trifolium_hirtum	NON-NATIVE CO~
644	PU-06	Peripheral Uplands	Avena_fatua	NON-NATIVE CO~
645	PU-06	Peripheral Uplands	Erodium_botrys	NON-NATIVE CO~
646	PU-06	Peripheral Uplands	Festuca_myuros	NON-NATIVE CO~
647	PU-06	Peripheral Uplands	Lactuca_serriola	NON-NATIVE CO~
648	PU-06	Peripheral Uplands	Lepidium_didymum	NON-NATIVE CO~
649	PU-06	Peripheral Uplands	Spergularia_sp	NON-NATIVE CO~
650	PU-06	Peripheral Uplands	Salsola_tragus	NON-NATIVE CO~
651	PU-06	Peripheral Uplands	Sonchus_sp	NON-NATIVE CO~
652	PU-06	Peripheral Uplands	Hordeum_murinum	NON-NATIVE CO~
653	PU-06	Peripheral Uplands	Bromus_catharticus	NON-NATIVE CO~
654	PU-06	Peripheral Uplands	Lythrum_hyssopifolia	NON-NATIVE CO~
655	PU-06	Peripheral Uplands	Sonchus_asper	NON-NATIVE CO~
656	PU-06	Peripheral Uplands	Erodium_cicutarium	NON-NATIVE CO~
657	PU-06	Peripheral Uplands	Festuca_bromoides	NON-NATIVE CO~
658	PU-06	Peripheral Uplands	Geranium_dissectum	NON-NATIVE CO~
659	PU-06	Peripheral Uplands	Trifolium_repens	NON-NATIVE CO~
660	PU-06	Peripheral Uplands	Eucalyptus_sp	NON-NATIVE CO~
661	PU-06	Peripheral Uplands	Centaurium_tenuiflorum	NON-NATIVE CO~
662	PU-06	Peripheral Uplands	Medicago_lupulina	NON-NATIVE CO~
663	PU-06	Peripheral Uplands	Eucalyptus_globulus	NON-NATIVE CO~
664	PU-06	Peripheral Uplands	Hordeum_marinum	NON-NATIVE CO~
665	PU-06	Peripheral Uplands	mulch/woodchips	OTHER COVER

666	PU-06	Peripheral Uplands	unlisted	OTHER COVER
667	PU-06	Peripheral Uplands	Natural Thatch	THATCH
668	PU-07	Peripheral Uplands	Bare Ground	BARE GROUND
669	PU-07	Peripheral Uplands	Eriogonum_parvifolium	NATIVE COVER
670	PU-07	Peripheral Uplands	Artemisia_californica	NATIVE COVER
671	PU-07	Peripheral Uplands	Erigeron_canadensis	NATIVE COVER
672	PU-07	Peripheral Uplands	Hazardia_squarrosa	NATIVE COVER
673	PU-07	Peripheral Uplands	Distichlis_spicata	NATIVE COVER
674	PU-07	Peripheral Uplands	Atriplex_lentiformis	NATIVE COVER
675	PU-07	Peripheral Uplands	Isocoma_menziesii	NATIVE COVER
676	PU-07	Peripheral Uplands	Bromus_diandrus	NON-NATIVE CO~
677	PU-07	Peripheral Uplands	Festuca_perennis	NON-NATIVE CO~
678	PU-07	Peripheral Uplands	Helminthotheca_echioides	NON-NATIVE CO~
679	PU-07	Peripheral Uplands	Medicago_polymorpha	NON-NATIVE CO~
680	PU-07	Peripheral Uplands	Melilotus_indicus	NON-NATIVE CO~
681	PU-07	Peripheral Uplands	Brachypodium_distachyon	NON-NATIVE CO~
682	PU-07	Peripheral Uplands	Brassica_nigra	NON-NATIVE CO~
683	PU-07	Peripheral Uplands	Bromus_hordeaceus	NON-NATIVE CO~
684	PU-07	Peripheral Uplands	Plantago_coronopus	NON-NATIVE CO~
685	PU-07	Peripheral Uplands	Sonchus_sp	NON-NATIVE CO~
686	PU-07	Peripheral Uplands	Atriplex_semibaccata	NON-NATIVE CO~
687	PU-07	Peripheral Uplands	Erodium_cicutarium	NON-NATIVE CO~
688	PU-07	Peripheral Uplands	Lepidium_didymum	NON-NATIVE CO~
689	PU-07	Peripheral Uplands	Polygonum_aviculare_depres~	NON-NATIVE CO~
690	PU-07	Peripheral Uplands	Spergularia_sp	NON-NATIVE CO~
691	PU-07	Peripheral Uplands	Hordeum_murinum	NON-NATIVE CO~
692	PU-07	Peripheral Uplands	Salsola_tragus	NON-NATIVE CO~
693	PU-07	Peripheral Uplands	Parapholis_incurva	NON-NATIVE CO~
694	PU-07	Peripheral Uplands	Polypogon_monspeliensis	NON-NATIVE CO~
695	PU-07	Peripheral Uplands	Washingtonia_robusta	NON-NATIVE CO~
696	PU-07	Peripheral Uplands	Hordeum_marinum	NON-NATIVE CO~
697	PU-07	Peripheral Uplands	Bromus_catharticus	NON-NATIVE CO~
698	PU-07	Peripheral Uplands	Sonchus_oleraceus	NON-NATIVE CO~
699	PU-07	Peripheral Uplands	mulch/woodchips	OTHER COVER
700	PU-07	Peripheral Uplands	unlisted	OTHER COVER
701	PU-07	Peripheral Uplands	Natural Thatch	THATCH

```
# confirmed cover_category values:
```

```
# "NATIVE COVER", "NON-NATIVE COVER", "BARE GROUND", "THATCH", "OTHER COVER"
```

```
# for all analyses we will filter to only NATIVE COVER and NON-NATIVE COVER, the rest would be
```

```
# confirmed clover species codes:
```

```
# sweet clover: Melilotus_indicus (and possibly Melilotus_albus (white sweet clover) - check
# bur clover: Medicago_polymorpha

# TODO: get exact transect names for fire-only, sheep-only, fire+sheep plots
```

```
melilotus_albus <- veg |>
  filter(psoc %in% c("Melilotus_albus",
                    "Melilotus_indicus"))
```